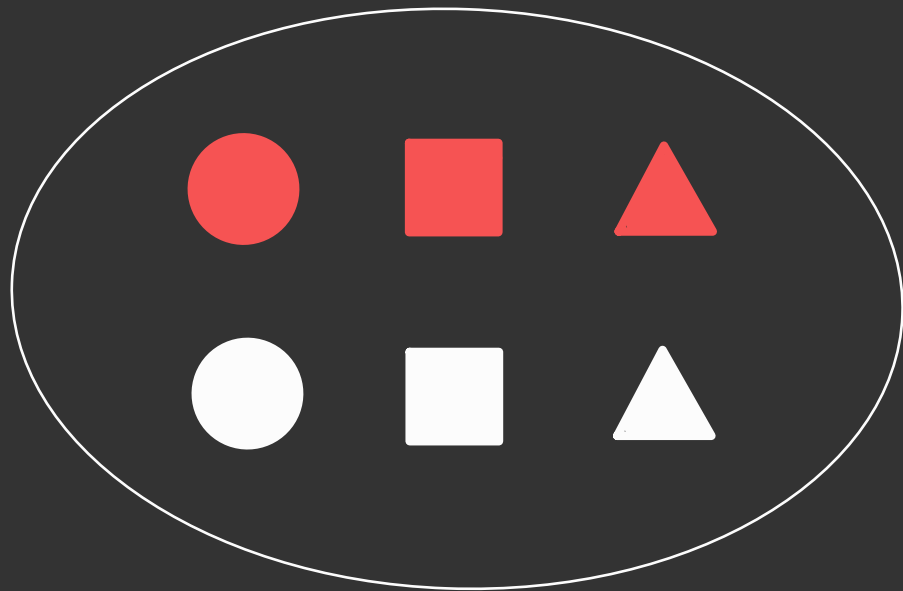


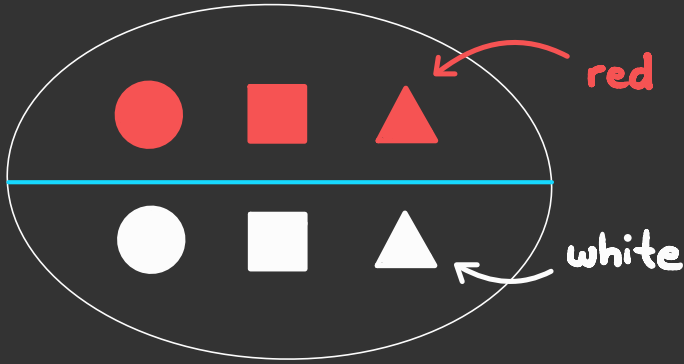
joint and marginal probability

CSCI
270

it can be helpful to use **variables**
to partition a sample space into regions



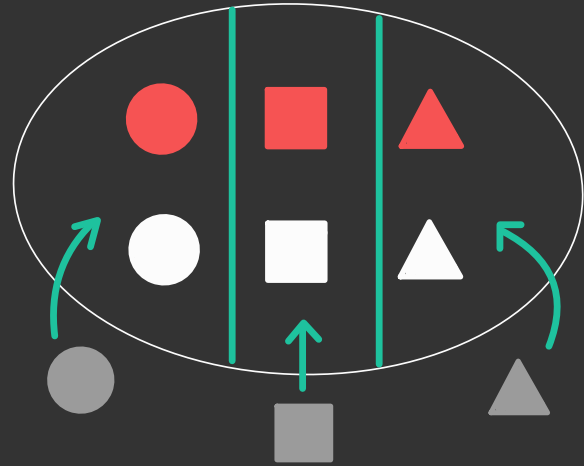
let **C** be the color
of the drawn card



$$\text{supp}(\text{C}) = \{\text{red}, \text{white}\}$$

"support"

let **S** be the shape
on the drawn card



$$\text{supp}(\text{S}) = \{\bullet, \blacksquare, \blacktriangle\}$$

"support"



ble**ssing** / misfortune

outcome

$P(\text{outcome})$



?

?

?

?

?

?



ble**ssing** / misfortune

outcome

$P(\text{outcome})$



$\frac{3}{25}$



$\frac{2}{25}$



$\frac{2}{25}$



$\frac{2}{25}$



$\frac{6}{25}$



$\frac{10}{25}$



ble**ssing** / misfortune

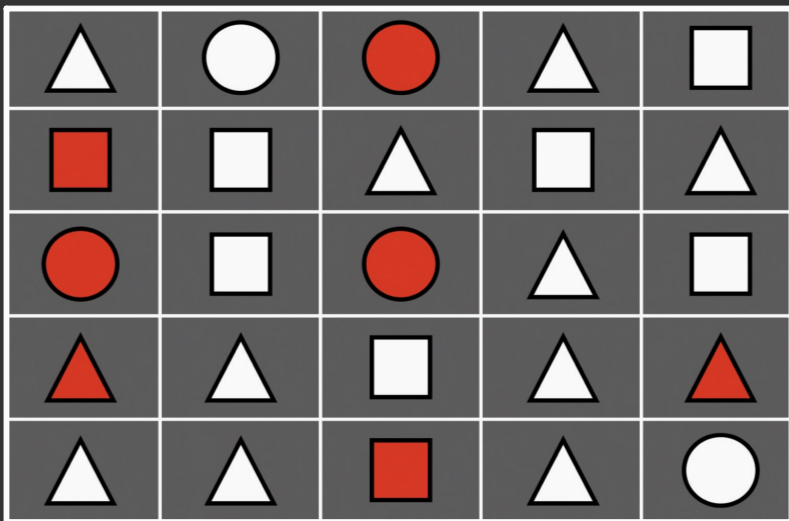
C	S	$P(c,s)$
red		$\frac{3}{25}$
red		$\frac{2}{25}$
red		$\frac{2}{25}$
white		$\frac{2}{25}$
white		$\frac{6}{25}$
white		$\frac{10}{25}$

joint distribution ↴



C	S	$P(c,s)$
red		$\frac{3}{25}$
red		$\frac{2}{25}$
red		$\frac{2}{25}$
white		$\frac{2}{25}$
white		$\frac{6}{25}$
white		$\frac{10}{25}$

what is the probability of misfortune?



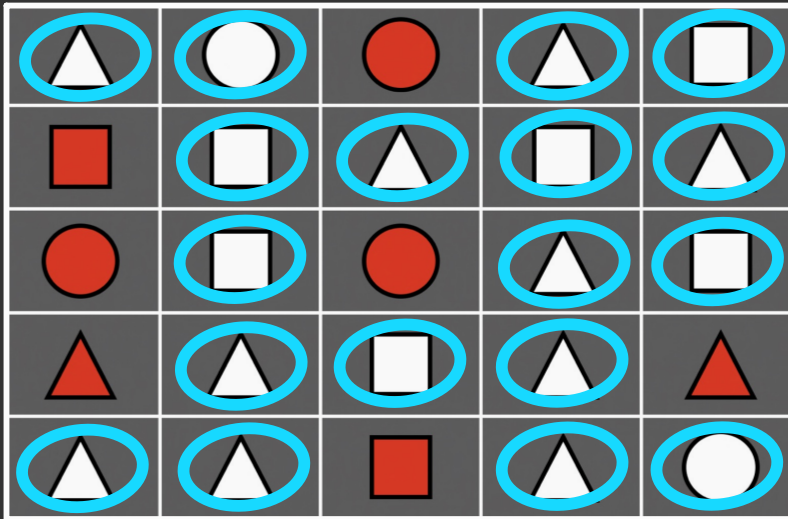
C	S	P(c,s)
red		$\frac{3}{25}$
red		$\frac{2}{25}$
red		$\frac{2}{25}$
white		$\frac{2}{25}$
white		$\frac{6}{25}$
white		$\frac{10}{25}$

what is $P(C = \text{white})$
~~the probability of misfortune~~ ?



C	S	$P(c,s)$
red		$\frac{3}{25}$
red		$\frac{2}{25}$
red		$\frac{2}{25}$
white		$\frac{2}{25}$
white		$\frac{6}{25}$
white		$\frac{10}{25}$

what is $P(C = \text{white})$?



C	S	$P(c,s)$
red		$\frac{3}{25}$
red		$\frac{2}{25}$
red		$\frac{2}{25}$
white		$\frac{2}{25}$
white		$\frac{6}{25}$
white		$\frac{10}{25}$

$$= \frac{18}{25}$$

joint probability

C	S	$P(c,s)$
red		$\frac{3}{25}$
red		$\frac{2}{25}$
red		$\frac{2}{25}$
white		$\frac{2}{25}$
white		$\frac{6}{25}$
white		$\frac{10}{25}$

$$= \frac{7}{25}$$

$$= \frac{18}{25}$$

marginal probability

C	$P(c)$
red	$\frac{7}{25}$
white	$\frac{18}{25}$

the law of total probability

(alternate version)

$$\underbrace{P(x_1, \dots, x_{n-1})}_{\text{marginal probability}} = \sum_{x_n \in \text{supp}(X_n)} \underbrace{P(x_1, \dots, x_n)}_{\text{joint probability}}$$

marginal
probability

C	P(c)
red	$\frac{7}{25}$
white	$\frac{18}{25}$

joint
probability

C	S	P(c,s)
red	●	$\frac{3}{25}$
red	■	$\frac{2}{25}$
red	▲	$\frac{2}{25}$
white	●	$\frac{2}{25}$
white	■	$\frac{6}{25}$
white	▲	$\frac{10}{25}$